



PDF Download
3696410.3714609.pdf
24 February 2026
Total Citations: 0
Total Downloads: 3769

Latest updates: <https://dl.acm.org/doi/10.1145/3696410.3714609>

RESEARCH-ARTICLE

Damage Analysis via Bidirectional Multi-Task Cascaded Multimodal Fusion

TAO LIANG, University of Electronic Science and Technology of China, Chengdu, Sichuan, China

SIYING WU, Southwest Jiaotong University, Chengdu, Sichuan, China

JUNFENG FANG, University of Science and Technology of China, Hefei, Anhui, China

GUOWU YANG, University of Electronic Science and Technology of China, Chengdu, Sichuan, China

WENYA WANG, Nanyang Technological University, Singapore City, Singapore

FENGMAO LV, Southwest Jiaotong University, Chengdu, Sichuan, China

Open Access Support provided by:

Nanyang Technological University

University of Electronic Science and Technology of China

Southwest Jiaotong University

University of Science and Technology of China

Published: 22 April 2025

[Citation in BibTeX format](#)

WWW '25: The ACM Web Conference 2025

April 28 - May 2, 2025
Sydney NSW, Australia

Conference Sponsors:
SIGWEB

Damage Analysis via Bidirectional Multi-Task Cascaded Multimodal Fusion

Tao Liang
University of Electronic Science and
Technology of China
Chengdu, China
Bytedance Inc.
Shenzhen, China
taoliangdpg@126.com

Siying Wu
Southwest Jiaotong University
Chengdu, China
siyingwuhere@126.com

Junfeng Fang
University of Science and Technology
of China
Hefei, China
fjf@mail.ustc.edu.cn

Guowu Yang
University of Electronic Science and
Technology of China
Chengdu, China
Kash Institute of Electronics And
Information Industry
Kashi, China
guowu@uestc.edu.cn

Wenya Wang
Nanyang Technological University
Singapore, Singapore
wangwy@ntu.edu.sg

Fengmao Lv*
Southwest Jiaotong University
Chengdu, China
fengmaolv@126.com

Abstract

Damage analysis in social media platforms such as Twitter is a comprehensive problem which involves different subtasks for mining damage-related information from tweets (e.g., informativeness, humanitarian categories and severity assessment). The comprehensive information obtained by damage analysis enables to identify breaking events around the world in real-time and hence provides aids in emergency responses. Recently, with the rapid development of web technologies, multimodal damage analysis has received increasing attentions due to users' preference of posting multimodal information in social media. Multimodal damage analysis leverages the associated image modality to improve the identification of damage-related information in social media. However, existing works on multimodal damage analysis address each damage-related subtask individually and do not consider their joint training mechanism. In this work, we propose the Bidirectional Multi-task Cascaded multimodal Fusion (BiMCF) approach towards joint multimodal damage analysis. To this end, we introduce the cascaded multimodal fusion framework to separately integrate effective visual and text information for each task, considering that different tasks attend to different information. To exploit the interactions across tasks, bidirectional propagation of the attended image-text interactive information is implemented between tasks, which can lead to enhanced multimodal fusion. Comprehensive experiments are

conducted to validate the effectiveness of the proposed approach. Code is available at <https://github.com/tiggers23/BiMCF>.

CCS Concepts

• **Human-centered computing** → **Social media**; • **Information systems** → **Multimedia information systems**.

Keywords

Damage analysis; social media Analysis; feature fusion; multimodal deep learning; multi-task learning.

ACM Reference Format:

Tao Liang, Siying Wu, Junfeng Fang, Guowu Yang, Wenya Wang, and Fengmao Lv. 2025. Damage Analysis via Bidirectional Multi-Task Cascaded Multimodal Fusion. In *Proceedings of the ACM Web Conference 2025 (WWW '25)*, April 28-May 2, 2025, Sydney, NSW, Australia. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3696410.3714609>

1 Introduction

Over the past years, social media platforms such as Twitter have become more and more popular in people's daily life. With the rapid development of web technologies, people tend to share their daily-life and news on social media in real-time. Recently, social medias have been exploited to identify breaking events around the world [1, 3, 8, 9, 37, 39], due to their increasing popularity. Event detection in social media can provide real-time emergency response to emergency events (e.g., natural disasters, vehicle damages and missing people) happened all around the world. In general, emergency event detection involves comprehensive damage analysis consisting of several subproblems, e.g., informativeness prediction, event classification and severity assessment [1].

The early works on social media damage analysis mainly focus on mining damage-related information solely from texts [5, 42]. Recently, multimodal damage analysis has received increasing attentions due to people's preference of posting multimodal information

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

WWW '25, Sydney, NSW, Australia

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-1274-6/25/04
<https://doi.org/10.1145/3696410.3714609>

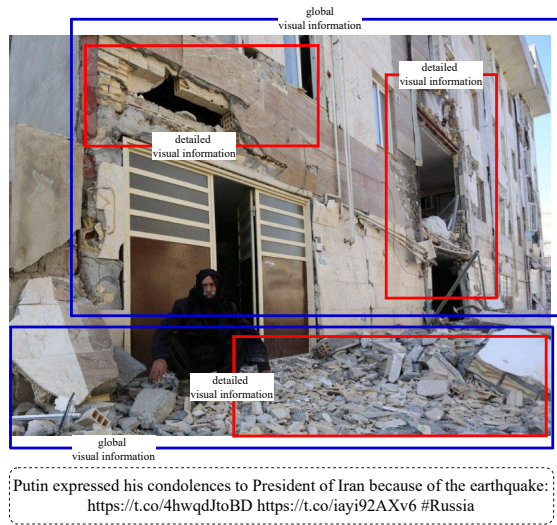


Figure 1: A damage analysis sample. The attended visual regions by different subtasks are marked with different colors. The blue box represents the informativeness subtask, while the red box represents the event classification subtask.

in social media [1, 4, 21]. Therefore, the CrisisMMD dataset is created as standard benchmark for multimodal damage analysis [4]. Samples of CrisisMMD are collected by crawling blogs posted on Twitter in times of seven natural disasters (*e.g.*, floods, wildfires, hurricanes, earthquakes) and featured by three class labels (*i.e.*, informativeness, humanitarian categories and severity). These class labels follow a cascaded structure, where only an informative post can be classified into a specific humanitarian category and only after this classification can the severity of the event be assessed. Afterwards, various multimodal learning methods are proposed based on the CrisisMMD benchmark [1, 2, 21, 23, 34]. Compared with unimodal methods, multimodal damage analysis leverages the associated image modality to improve the identification of damage-related information in social media [22].

However, most of existing works on multimodal damage analysis address each damage-related subtask independently and do not consider to leverage the inherent relationships across tasks [8, 28, 30, 38]. From the perspective of multi-task learning [26, 29, 51], a joint training framework can lead to improved performance for each task by exploiting the relationships between tasks. Although a multi-task damage assessment system has been designed in [2], it only organizes subtasks in a pipelined way and does not consider the joint training of them. The motivation of jointly training damage analysis subtasks establishes the multimodal multi-task learning framework. Similar to general multi-task learning methods [51, 53], existing multimodal multi-task methods mainly focus on learning shared multimodal representations [15, 35, 50], which effectively facilitate information sharing between different tasks. However, they only implicitly reflect task correlations, making it less effective when there are clear relationships among tasks. Figure 1 displays a multimodal damage analysis sample about two cascaded tasks and we can observe that informativeness prediction

and event classification subtasks attend to different image-text interactive information. Specifically, the informativeness prediction task pays attention to global features, while the event classification task further narrows down to specific detail features based on the attended information by the informativeness prediction task. The shared multimodal representations can’t explicitly explore such intrinsic interdependencies, thus leading to degraded performance.

In this work, we propose the Bidirectional Multi-Task Cascaded multimodal Fusion (BiMCF) approach towards joint multimodal damage analysis. Different from previous works which implement multi-task learning based on shared multimodal representations, our approach deploys the multi-task interaction mechanism in the implementation of multimodal fusion. For different tasks, we introduce the cascaded multimodal fusion framework to separately integrate visual and text information for each task, considering the fact that different tasks attend to different information. Furthermore, the forward and backward gate mechanisms are introduced in our bidirectional propagation framework for each task to enhance task-specific feature extraction and facilitate effective multimodal fusion. Specifically, the forward gate concentrates on amplifying salient image features to provide more detailed information, while the backward gate focuses on expanding attention regions to offer richer global information. Therefore, our bidirectional cascaded structure can alleviate errors accumulated in one direction and enhance the identification of damage-related information. Unlike the prior works [6, 27, 45] which consider multi-task cascades in the final prediction of labels, our approach deploys the cascaded structure of tasks into the multimodal fusion process of each task, thereby explicitly encoding the interdependencies among subtasks into the model structure. To this end, the cross-modal attention mechanism [1, 12, 13, 17, 48] incorporates information from both textual and visual modalities more effectively.

To sum up, the contributions of this work are mainly three-fold:

- We propose to implement comprehensive multimodal damage analysis from the multi-task learning perspective, with the purpose of exploiting the potential relationships between tasks.
- We propose the BiMCF approach by introducing the bidirectional cascaded framework to implement multimodal multi-task learning, which encodes the interdependencies among subtasks into the model structure and enhance the multimodal fusion process.
- Comprehensive experiments are conducted to show the effectiveness of our approach. Our approach can achieve better performance than existing works.

2 Related Work

2.1 Damage Analysis

Damage analysis in social media can provide real-time response to emergency events. Previous works on damage analysis mainly focus on mining damage-related information solely from images [20, 33] or texts [5, 42, 43]. Recently, attention has been paid to multimodal damage analysis due to people’s increasing interests in posting multimodal messages on social media platforms. In particular, Alam et al. [4] create the CrisisMMD dataset as standard benchmark for multimodal damage analysis by crawling blogs from Twitter. Agarwal et al. [2] propose the Crisis-DIAS framework towards comprehensive multimodal damage analysis based on CrisisMMD.

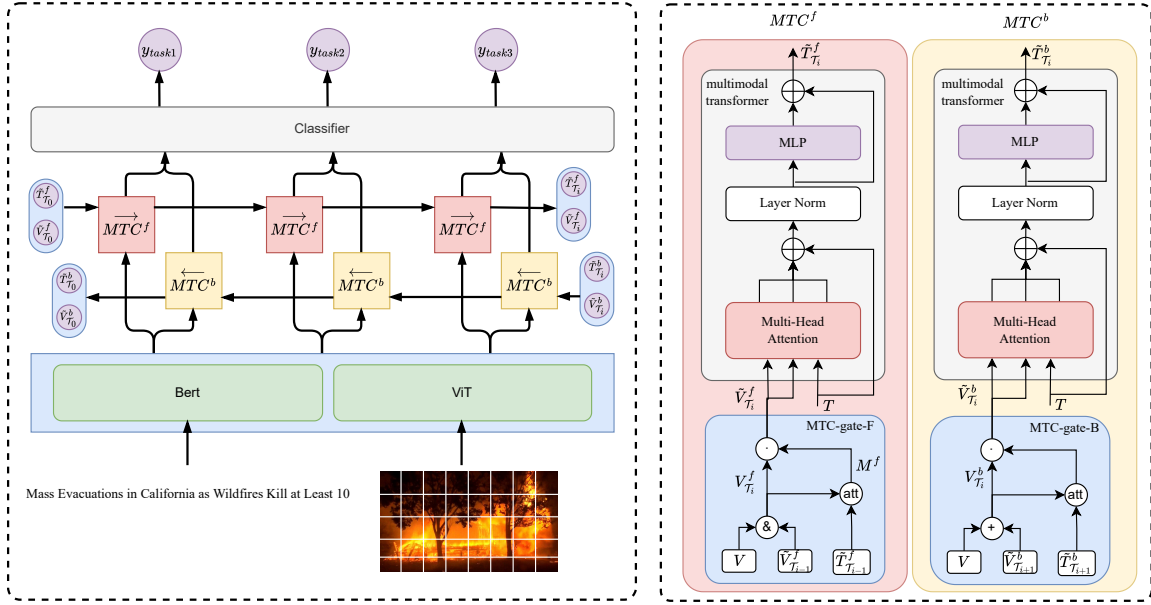


Figure 2: (Left) The overall architecture of the proposed BiMCF approach. First, for a pair of text and image input, we use BERT and ViT to generate feature representations respectively. Then the image and text features are passed and updated through MTC^f and MTC^b . Last, the final prediction for each task $\mathcal{T}_i$ is decided by two output text features from both directions, namely $T_{\mathcal{T}_i}^b$ and $T_{\mathcal{T}_i}^f$. (Right) The detailed frameworks of MTC^f and MTC^b . ‘&’ demotes max operator and ‘+’ denotes addition operator.

Recently, Liang et al. [23] consider fine-grained cross-modal interactions in feature fusion and obtain improved damage categorization performance. Furthermore, Bishwas et al. [28] propose to fine-tune pre-trained contrastive models to distinguish multimodal crisis tweets. However, existing works address each damage-related sub-task individually and do not consider their joint training mechanism.

2.2 Multimodal Multi-Task Learning

Multi-task learning exploits task relationships to regularize the training of deep models. Most of existing works implement multi-task learning based on task-shared representations. In multimodal settings, existing works also build on task-shared multimodal representations [27, 31, 45, 50], where features of different modalities are firstly fused into a multimodal representation which is then utilized to implement multi-task learning. Following this framework, Tan et al. [45] propose the MultiCoFusion model for cancer prognosis prediction from multi-modal inputs. Their approach performs alternate training between different tasks on task-shared multimodal representations. Alam et al. [6] proposes the Multimodal Spatiotemporal Neural Fusion network for Multi-Task Learning, in which multi-task cascaded learning are developed on the fusion layer. Maity et al. [27] implement both FeedBack Multi-task learning and Central-Net Multi-task learning on fused features. Zhang et al. [50] propose to share multimodal features on a sparse space, ensuring that only features beneficial to decision-making are shared. In general, existing works do not fully consider the uniqueness and correlation of different tasks during multimodal fusion. Different from existing works which only leverage task relationships for final predictions, this work further considers to exploit multi-task interactions in the implementation of multimodal fusion and introduces a more reasonable multi-task cascaded multimodal fusion framework.

3 METHODOLOGY

3.1 Problem Definition

In general, damage analysis involves three related subtasks [1, 2, 4, 23], namely informativeness prediction, event classification and severity prediction. We adopt the same label settings used in [23]. In particular, the informativeness prediction task is to determine whether the post is useful for humanitarian aid purpose. The event classification task is to further identify the damage types of the informative post. The severity prediction task is to assess the severity level of the identified damage. The above three tasks follow the cascaded relationship, where only an informative post can be classified into a specific humanitarian category and only after this classification can the severity of the event be assessed.

In a multimodal setting, we are given a pair of text and image input $X = \{T, V\}$ where T denotes a text sequence, and V denotes an image. The outputs consist of three labels $Y = \{y_I, y_H, y_S\}$ where y_I is labeled as *informative/not informative/don't know or can't judge*, y_H is labeled as *affected individuals (injury, dead, missing, found, etc.) / infrastructure and utility damage / rescue, volunteering or donation efforts / vehicle damage / other relevant information*, and y_S is labeled as *severe damage / mild damage / little or no damage / don't know or can't judge*. We denote the training data consisting of both paired text and image input as well as a label set for all the three subtasks as $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^N$. During inference, the target is to predict the set of labels Y_j for each test instance X_j .

3.2 Motivation & Model Framework

Multi-task frameworks have been widely used to solve problems involving multiple tasks in a joint manner. These frameworks include pipelined approaches [19, 46] in which some tasks are performed in

the first place and their outputs are passed to the remaining tasks. This operation results in error propagation when the model does not perform well on earlier tasks. Another strategy adopts parameter sharing [29, 41] among different tasks with the intuition that the shared parameters encode common features across tasks. However, parameter sharing only implicitly reflects task correlations, making it less effective when there are clear relationships among subtasks, which is the case for multi-task damage analysis.

To explicitly infuse the evolution of the involved tasks, we propose a bidirectional multi-task cascaded framework which ranks the three subtasks in the order of informativeness prediction, humanitarian categorization and severity level prediction based on the granularity (coarse to fine) of the problem. In the three cascaded tasks, the later task further narrows down to specific detailed features based on the attended information by its preceding task. Inspired from the working mechanism of a zoom camera which narrows the range of the lens but amplifies the focused area each time it zooms in, we treat the ordered task cascade as a zooming process. In a forward pass, we sequentially zoom in images based on the output from a preceding task. On the other hand, an addition of a backward pass performs the opposite operation which zooms out an image given the output from a finer-grained task. This bidirectional cascade structure propagates information in dual directions to alleviate errors accumulated in one direction. Meanwhile, the bidirectional flows effectively exchange information from related tasks to enhance the prediction for the target subtask.

To incorporate information from both textual and visual modalities, we further adopt a unified multimodal fusion framework across all the tasks. Specifically, we treat text representations as a common reference which is used to guide the attention computations over the resulting image representations at each zooming step. The final multimodal features for each task are obtained via cross-modal transformers. Using text modality as a reference is attributed to the fact that information is usually more explicitly expressed in natural languages. To this end, we propose a bidirectional multi-task cascaded framework with multimodal fusion to solve the three subtasks of damage analysis in an end-to-end manner.

The overall architecture of BiMCF is shown in Figure 2. Specifically, given an input textual description T and an image V , we first generate a feature representation for each token in the text as $\mathbf{T} = [\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_n]$ using a pretrained language model, e.g., BERT [10]. The image feature representations are produced following the existing works using ViT [11]. In particular, the image is first resized to 224×224 and then divided into m image patches which are further reshaped into a sequence. We extract the features from the last layer of ViT as visual input representations, denoted as $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m]$. Here we define $\mathbf{T} \in \mathbb{R}^{n \times D_t}$ and $\mathbf{V} \in \mathbb{R}^{m \times D_v}$. BiMCF processes the input features $\mathbf{T}$ and $\mathbf{V}$ in two opposite directions, both consisting of three modules corresponding to three subtasks. Each i -th module incorporates a Multimodal Task Cascading (MTC) gate which takes the original image features $\mathbf{V}$, the output image features $\tilde{\mathbf{V}}_{\mathcal{T}_{i-1}}$ and the output text features $\tilde{\mathbf{T}}_{\mathcal{T}_{i-1}}$ from its preceding task as input to decide the information flow from its preceding task. We design different gate operations for forward pass (denoted as $\mathcal{G}_{MTC-f}$) and backward pass (denoted as $\mathcal{G}_{MTC-b}$). The output $\tilde{\mathbf{V}}_{\mathcal{T}_i}$ from each MTC gate is then treated

as the updated image representation to be fused with the text representation via a multimodal transformer block. We use the final output text features $\tilde{\mathbf{T}}_{\mathcal{T}_i}$ after the multimodal fusion process for task predictions. The details of each component will be further illustrated in the following sections.

3.3 Task-Specific Multimodal Transformers

For each task, we exploit the complex interactions between the textual modality and the visual modality using multimodal transformers. Specifically, for each task $\mathcal{T}_i$, given input features for a sentence $\mathbf{T} = [\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]$ and an image $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_m]$, the multimodal transformers construct its query matrix using the text features $\mathbf{T}\mathbf{W}_q$. The key and value matrices are generated based on image features $\mathbf{V}\mathbf{W}_k$ and $\mathbf{V}\mathbf{W}_v$, respectively, where $\mathbf{W}_q \in \mathbb{R}^{D_t \times D}$, $\mathbf{W}_k, \mathbf{W}_v \in \mathbb{R}^{D_v \times D}$ are transformation matrices. The cross-modal attentions then compute the attention weights over image patches and generate an updated output matrix as:

$$\mathbf{H} = \text{softmax} \left(\frac{(\mathbf{T}\mathbf{W}_q)(\mathbf{V}\mathbf{W}_k)^\top}{\sqrt{D}} \right) (\mathbf{V}\mathbf{W}_v). \quad (1)$$

With multiple heads, the final output text representation, denoted as $\tilde{\mathbf{T}}$, is computed as

$$\tilde{\mathbf{T}} = f_{LN}(f_{FF}([\mathbf{H}_1; \dots; \mathbf{H}_M]) + \mathbf{T}). \quad (2)$$

Here f_{LN} and f_{FF} denote layer norm and feed-forward functions, respectively. M denotes the number of attention head. We use different transformer parameters for different subtasks and denote by $\tilde{\mathbf{T}}_{\mathcal{T}_i} = \mathcal{M}_i(\mathbf{V}, \mathbf{T})$ as the multimodal transformer function for task $\mathcal{T}_i$ obtained through equation (1) and (2).

3.4 Multimodal Task Cascade

Task-specific multimodal transformers only exploit cross-modal interactions and ignore the essential relationships among different tasks. In this aspect, we target on multi-task setting with a cascaded structure and explicitly encode task relationships into the model architecture via Multimodal Task Cascading (MTC) gate. To capture their dependencies, we use MTC gates within each task module to explicitly update the features of the input image. Different from common practices that use the same image input for multiple tasks, we feed different tasks with distinct image features generated from the MTC gate. In what follows, we introduce two gate mechanisms adopted in forward flow and backward flow, respectively.

Forward Gate. In the forward flow, we order the three tasks according to the increasing granularity level, namely informativeness prediction, humanitarian categorization, followed by severity prediction. The later task requires more fine-grained analysis of the input than its preceding task and could benefit from the features of its preceding task. For example, suppose task $\mathcal{T}_i$ aims to predict whether there is an event and is provided with a scaled image focusing on the upper right corner. When the subsequent task $\mathcal{T}_{i+1}$ aims to predict the exact event type, it could be beneficial to zoom in the upper right corner of the image to look for more salient evidence, because the rest parts of the image are not relevant to events. Hence, the objective of the forward gate is to further amplify the salient features in each image based on the selected regions generated in the

preceding task. This is similar to the zoom-in process of a camera which narrows down to specific areas for a deeper examination.

Formally, for task $\mathcal{T}_i$, the forward gate considers its interaction with the preceding task $\mathcal{T}_{i-1}$. Given the original image features $\mathbf{V}$, the output text features $\tilde{\mathbf{T}}_{\mathcal{T}_{i-1}}^f$ and image features $\tilde{\mathbf{V}}_{\mathcal{T}_{i-1}}^f$ computed from the module of task $\mathcal{T}_{i-1}$ in the forward pass, the forward MTC gate computes a weight score s_j^f for each image patch to scale its effect prior to the multimodal fusion process. Here we use the superscript f to denote the forward pass. Specifically, we first apply a max operator over two image features $\mathbf{V}$ and $\tilde{\mathbf{V}}_{\mathcal{T}_{i-1}}^f$,

resulting in $\mathbf{V}_{\mathcal{T}_i}^f = \max(\mathbf{V}, \tilde{\mathbf{V}}_{\mathcal{T}_{i-1}}^f)$, where $\max$ is an element-wise selector. This operation mimics the zoom-in effect which tries to keep salient features obtained in the preceding task. On the other hand, the preceding task may propagate errors that negatively affect the current task. The max operator could mitigate this issue by retaining salient features from the original image. The second step computes a scaling score for each image patch according to:

$$\mathbf{M}^f = \mathcal{L}_1^f(\mathbf{V}_{\mathcal{T}_i}^f) \cdot \mathcal{L}_2^f(\tilde{\mathbf{T}}_{\mathcal{T}_{i-1}}^f)^\top, \quad (3)$$

$$s_j^f = \frac{\exp^{\sum_{k=1}^n \mathbf{M}_{[j,k]}^f}}{\sum_{j'=1}^m \exp^{\sum_{k=1}^n \mathbf{M}_{[j',k]}^f}}, \quad (4)$$

where $\mathcal{L}_1^f$ and $\mathcal{L}_2^f$ denote two linear transformation functions used to project the inputs $\mathbf{V}_{\mathcal{T}_i}^f$ and $\tilde{\mathbf{T}}_{\mathcal{T}_{i-1}}^f$ into $\mathbb{R}^{m \times D_v}$ and $\mathbb{R}^{n \times D_v}$, respectively. Each element $\mathbf{M}_{[j,k]}^f$ in $\mathbf{M}^f \in \mathbb{R}^{m \times n}$ stores the correlation score between the j -th image patch and the k -th text token. This reflects cross-modality correlations and treats the text modality as a reference to guide the zooming process. Equation (4) applies a softmax operation over the sum of correlation scores of each image patch with text tokens. A larger score implies higher correlations of its corresponding image patch with the text input, and thus implies higher chance of being retained for the target task. The final output image features are

$$\tilde{\mathbf{V}}_{\mathcal{T}_i}^f = \mathbf{s}^f \odot \mathbf{V}_{\mathcal{T}_i}^f, \quad (5)$$

where $\odot$ denotes row-wise multiplications. An exception happens to the first subtask $\mathcal{T}_1$ where there is no preceding task for guidance. In this case, we use $\tilde{\mathbf{T}}_{\mathcal{T}_0}^f = \mathbf{T}$ and $\tilde{\mathbf{V}}_{\mathcal{T}_0}^f = \mathbf{V}$ as the input to the forward gate. For ease of notation, we use $\tilde{\mathbf{V}}_{\mathcal{T}_i}^f = \mathcal{G}_{\text{MTC-f}}(\mathbf{V}, \tilde{\mathbf{T}}_{\mathcal{T}_{i-1}}^f, \tilde{\mathbf{V}}_{\mathcal{T}_{i-1}}^f)$ to denote all the above-mentioned computations for the forward gate. **Backward Gate.** In the backward flow, we starts from the most fine-grained subtask, *i.e.*, severity level prediction, followed by humanitarian categorization and informativeness prediction. Generally, the finer-grained task provides a small range of focused image area that is most effective for predictions. It is then beneficial to start from this precise area and gradually expand the region around it for subsequent more coarse-grained tasks. This assembles the zoom-out process of a camera which helps to expand the search space. Hence, the objective of the backward gate is to pivot image attentions around anchor regions obtained from the preceding task.

Formally, for task $\mathcal{T}_i$, the backward gate $\mathcal{G}_{\text{MTC-b}}$ considers the interaction of $\mathcal{T}_i$ and $\mathcal{T}_{i+1}$. The input of the backward gate consists of the original image features $\mathbf{V}$, the output text features $\tilde{\mathbf{T}}_{\mathcal{T}_{i+1}}^b$ and

image features $\tilde{\mathbf{V}}_{\mathcal{T}_{i+1}}^b$ from task $\mathcal{T}_{i+1}$, in which the superscript b denotes the backward pass. Different from the forward gate, the first step of the zoom-out process uses element-wise addition to simulate the expanding effect: $\mathbf{V}_{\mathcal{T}_i}^b = \mathbf{V} + \tilde{\mathbf{V}}_{\mathcal{T}_{i+1}}^b$. Similar to the forward gate, the second step then computes the scaling score s^b as follows:

$$\mathbf{M}^b = \mathcal{L}_1^b(\mathbf{V}_{\mathcal{T}_i}^b) \cdot \mathcal{L}_2^b(\tilde{\mathbf{T}}_{\mathcal{T}_{i+1}}^b)^\top, \quad (6)$$

$$s_j^b = \frac{\exp^{\sum_{k=1}^n \mathbf{M}_{[j,k]}^b}}{\sum_{j'=1}^m \exp^{\sum_{k=1}^n \mathbf{M}_{[j',k]}^b}}. \quad (7)$$

The last step produces the final image representation during the backward propagation:

$$\tilde{\mathbf{V}}_{\mathcal{T}_i}^b = \mathbf{s}^b \odot \mathbf{V}_{\mathcal{T}_i}^b. \quad (8)$$

Similarly, we use $\tilde{\mathbf{V}}_{\mathcal{T}_i}^b = \mathcal{G}_{\text{MTC-b}}(\mathbf{V}, \tilde{\mathbf{T}}_{\mathcal{T}_{i+1}}^b, \tilde{\mathbf{V}}_{\mathcal{T}_{i+1}}^b)$ to denote all computations for the backward MTC gate. When the task corresponds to the third subtask $\mathcal{T}_3$, we use $\tilde{\mathbf{T}}_{\mathcal{T}_4}^b = \mathbf{T}$ and $\tilde{\mathbf{V}}_{\mathcal{T}_4}^b = \mathbf{V}$ as the input.

3.5 Joint Multi-Task Multimodal Learning

To jointly learn the three cascaded tasks in a multimodal setting, we propose a Bidirectional Multi-task Cascaded multimodal Fusion framework integrating the task-specific multimodal transformers and the cascading gate respectively introduced in Section 3.3 and 3.4. Specifically, the model takes a pair of text-image $(\mathbf{T}, \mathbf{V})$ as the input to all three task modules. In a forward pass, we first use the forward MTC gate to produce an output image representation for task $\mathcal{T}_1$: $\tilde{\mathbf{V}}_{\mathcal{T}_1}^f = \mathcal{G}_{\text{MTC-f}}(\mathbf{V}, \mathbf{T}, \mathbf{V})$. Then a multimodal transformer block is applied on top of $\tilde{\mathbf{V}}_{\mathcal{T}_1}^f$ and the input text features $\mathbf{T}$ to obtain the

output text representation corresponding to $\mathcal{T}_1$: $\tilde{\mathbf{T}}_{\mathcal{T}_1}^f = \mathcal{M}_1(\tilde{\mathbf{V}}_{\mathcal{T}_1}^f, \mathbf{T})$. Subsequently, we obtain outputs for $\mathcal{T}_i$ ($i = 2, 3$) sequentially via $\tilde{\mathbf{V}}_{\mathcal{T}_i}^f = \mathcal{G}_{\text{MTC-f}}(\mathbf{V}, \tilde{\mathbf{T}}_{\mathcal{T}_{i-1}}^f, \tilde{\mathbf{V}}_{\mathcal{T}_{i-1}}^f)$ and $\tilde{\mathbf{T}}_{\mathcal{T}_i}^f = \mathcal{M}_i(\tilde{\mathbf{V}}_{\mathcal{T}_i}^f, \mathbf{T})$.

In the backward pass, we first apply the backward MTC gate on the most fine-grained task $\mathcal{T}_3$ to produce an output image representation: $\tilde{\mathbf{V}}_{\mathcal{T}_3}^b = \mathcal{G}_{\text{MTC-b}}(\mathbf{V}, \mathbf{T}, \mathbf{V})$. Then a multimodal transformer block is applied on top of $\tilde{\mathbf{V}}_{\mathcal{T}_3}^b$ and the input text features $\mathbf{T}$ to obtain the output text representation corresponding to $\mathcal{T}_3$: $\tilde{\mathbf{T}}_{\mathcal{T}_3}^b = \mathcal{M}_3(\tilde{\mathbf{V}}_{\mathcal{T}_3}^b, \mathbf{T})$. Subsequently, we obtain outputs for $\mathcal{T}_i$ ($i = 2, 1$) sequentially via $\tilde{\mathbf{V}}_{\mathcal{T}_i}^b = \mathcal{G}_{\text{MTC-b}}(\mathbf{V}, \tilde{\mathbf{T}}_{\mathcal{T}_{i+1}}^b, \tilde{\mathbf{V}}_{\mathcal{T}_{i+1}}^b)$ and $\tilde{\mathbf{T}}_{\mathcal{T}_i}^b = \mathcal{M}_i(\tilde{\mathbf{V}}_{\mathcal{T}_i}^b, \mathbf{T})$. The final prediction for each task $\mathcal{T}_i$ is made by concatenating the output text features from both directions:

$$\tilde{y}_i = \mathcal{F}([\tilde{\mathbf{T}}_{\mathcal{T}_i}^f; \tilde{\mathbf{T}}_{\mathcal{T}_i}^b]). \quad (9)$$

Here $\mathcal{F}$ indicates two linear layers.

During training, the loss of $\mathcal{T}_1$ is:

$$L_{\mathcal{D}_1} = \frac{1}{N_1} \sum_{j=1}^{N_1} L_{\mathcal{T}_1}(X_j, y_j), \quad (10)$$

where each $L_{\mathcal{T}_1}(X_j, y_j)$ computes a cross-entropy loss between the prediction $\tilde{y}_j$ and the ground-truth label y_j for each instance, and N_1 denotes the total number of training samples in $\mathcal{T}_1$.

Similarly, the losses of $\mathcal{T}_2$ and $\mathcal{T}_3$ are:

$$L_{\mathcal{D}_2} = \frac{1}{N_2} \sum_{j=1}^{N_2} L_{\mathcal{T}_2}(X_j, y_j), \quad (11)$$

$$L_{\mathcal{D}_3} = \frac{1}{N_3} \sum_{j=1}^{N_3} L_{\mathcal{T}_3}(X_j, y_j). \quad (12)$$

We further introduce three hyper-parameters λ , γ and σ , considering that each task contributes differently to the final outcome. Thus, our objective is to minimize the following aggregated loss:

$$L_{\mathcal{D}} = \lambda L_{\mathcal{D}_1} + \gamma L_{\mathcal{D}_2} + \sigma L_{\mathcal{D}_3}. \quad (13)$$

4 Experiments

In this section, we design comprehensive experiments to answer the following four research questions:

- **RQ1:** How does our proposed BiMCF perform on damage analysis compared to other baselines? Why does our BiMCF arrange subtasks from coarse to fine granularity?
- **RQ2:** Can the MTC forward gate and backward gate achieve zoom-in and zoom-out effects respectively? Do they facilitate information complementarity?
- **RQ3:** How does our bidirectional propagation mechanism perform compared to unidirectional propagation?
- **RQ4:** Does our proposed BiMCF demonstrate strong robustness?

We also analyze the sensitivity of our BiMCF to hyper-parameters and the layer number of bidirectional flow in the appendix. Moreover, the generalization of our method is discussed in the appendix.

4.1 Implementation Details

Datasets. Following the prior works on multimodal damage analysis [1, 2, 21, 23], we adopt the standard CrisisMMD benchmark to validate our approach. The samples in this dataset are multimodal social media posts consisting of image-text pairs, which are crawled from Twitter in times of seven natural disasters (e.g., floods, wildfires, hurricanes, earthquakes). Moreover, this benchmark consists of three cascaded subtasks, i.e., informativeness prediction (task1), events classification (task2), severity prediction (task3). The class label of the latter task depends on that of the former task.

Settings. We employ pre-trained BERT-base and ViT as our language and image backbones to generate 768-dimensional features. The image input is first resized to 224×224 and then divided into 16×16 patches. During the training process, Adam is adopted as the optimizer with a fixed learning rate of 0.000015. We use a batch size of 32 and train the model for 20 epochs. Moreover, the dropout rate is set as 0.8 and the hyper-parameters are determined by the validation set. All the experiments are run on four A100 GPUs.

Baselines. The proposed approach is compared with several existing state-of-the-art baselines, including Cross-Attention [1], Crisis-DIAS [2], interactive multi-task learning network (IMN) [14], Aspect-oriented Method (AoM) [52], Dual Query Prompt Sentiment Analysis (DQPSA) [36], multimodal joint learning (JML) [17], VL-BERT [44], Vision and Language Transformer (ViLT) [18], Qwen-VL [7] and CogVLM [47]. Among them, Cross-Attention filters useless information to achieve effective multimodality fusion. AoM, DQPSA, Crisis-DIAS, IMN and JML are specifically designed for multimodal

multi-task learning. VL-BERT and ViLT are typically pretrained vision and language models, which can quickly adapt to downstream tasks by fine-tuning. Two multimodal large language models (MLLMs), Qwen-VL and CogVLM, are also incorporated.

4.2 Performance Comparison (RQ1)

We compare our BiMCF with the above baselines and present the results on the CrisisMMD in Table 1. Additionally, we conduct experiments across all possible task sequences to demonstrate the superiority of our task sequence and display the results in Table 2.

- **Obs1: Our proposed BiMCF achieves state-of-the-art results across all the adopted baselines, demonstrating satisfactory performance improvements compare to both single_task approaches and multi_task approaches.** From Table 1, we can draw the following observations. First, we find that the same model generally performs better in the single_task setting than in the parallel multi_task setting, suggesting that parallel subtasks can negatively interfere with each other. Although existing cascaded multi_task methods take the task interdependencies into account, they still fail to outperform our approach. This can be attributed to that our BiMCF fully explores the underlying relationships between the cascaded tasks and the bidirectional flows effectively suppress the propagation of negative messages, thus enabling a further augmentation of the multimodal fusion.
- **Obs2: Our proposed BiMCF exhibits superior capability in disaster analysis compared to MLLMs.** When utilizing text prompts to enable Qwen-VL and CogVLM to perform damage predictions, we can observe from Table 1 that there remains an obvious performance gap between the two MLLMs and ours, due to the lack of domain-specific knowledge in disaster analysis and the insufficient exploration of subtask relationships. Similarly, MLLMs fine-tuned with LoRA [16] also fail to explore their cascading relationships and achieve inferior performance to ours.
- **Obs3: Arranging the three subtasks from coarse to fine granularity outperforms other possible task sequences.** In Table 2, our sequential approach achieves the best performance, with an average improvement of 2.1% to 4.9% across all tasks. This is because our adopted sequence facilitates a gradual understanding of complex information and seamlessly aligns with the intrinsic interdependencies among the three subtasks.

4.3 Visualization (RQ2)

In this part, we present the visualization of the attended image features in different subtasks and pass mechanisms in Figure 3.

- **Obs4: The MTC forward gate achieves the zoom-in effect to keep salient features, while the MTC backward gate achieves the zoom-out effect to obtain rich global information.** In the forward flow, we can observe from top to bottom that the attended image regions are progressively narrowed down to specific areas. This is because later tasks are more fine-grained and focus more on specific details. Similarly, in the backward flow, we can observe from bottom to up that the attended areas expand around some anchor regions. This is because preceding tasks are more coarse-grained and require abundant global information.
- **Obs5: The combined MTC forward & backward gate can comprehensively locate task-specific features and avoid**

Table 1: Performance comparison with baselines on CrisisMMD. ‘Single_task’ denotes treating each subtask independently. ‘Parallel multi_task’ and ‘Cascaded multi_task’ address all three subtasks simultaneously. However, the former doesn’t consider the inherent dependencies between subtasks, while the latter does. The best results are in bold, and the second-best results are underlined. Metrics include Acc: classification accuracy, M-F1: Macro F1-score and W-F1: weighted F1-score.

Category	Method	Task 1			Task 2			Task 3		
		Acc(%)	M-F1(%)	W-F1(%)	Acc(%)	M-F1(%)	W-F1(%)	Acc(%)	M-F1(%)	W-F1(%)
Single_task	Cross-Attention [1]	83.7	83.1	83.6	80.7	74.6	81.1	88.9	51.6	88.5
	VLBert [44]	<u>84.1</u>	<u>83.9</u>	<u>84.0</u>	81.7	75.2	<u>81.9</u>	90.9	55.7	90.3
	ViLT [18]	83.6	83.3	83.4	<u>82.1</u>	74.7	<u>81.9</u>	90.2	55.3	89.7
	Qwen-VL [7]	75.2	74.6	75.1	71.7	69.2	71.5	86.8	50.8	86.6
	Qwen-VL+LoRA [16]	76.3	75.9	76.2	73.6	71.3	73.8	88.1	51.6	87.9
	CogVLM [47]	77.3	76.7	77.1	70.6	67.4	69.8	87.4	54.6	86.1
	CogVLM+LoRA	79.1	79.0	78.9	72.4	68.2	71.3	89.1	55.9	88.6
Parallel Multi_task										
Multi_task	Cross-Attention	80.7	80.1	80.4	79.7	72.3	79.1	86.9	50.8	85.9
	VLBert	81.9	81.3	81.7	80.7	72.7	80.2	89.2	54.1	89.0
	ViLT	81.2	80.9	81.0	80.1	72.6	79.8	88.7	54.6	88.1
	Qwen-VL	72.6	72.3	72.5	70.1	68.3	69.7	84.2	51.9	83.6
	Qwen-VL+LoRA	80.3	80.4	80.3	76.2	70.7	76.5	88.5	53.8	88.2
	CogVLM	75.3	75.1	75.3	68.1	75.4	67.7	85.5	53.6	85.1
	CogVLM+LoRA	81.9	81.7	81.8	78.9	76.1	78.5	89.6	54.1	89.4
	IMN [14]	81.8	81.5	81.8	80.7	<u>77.1</u>	80.4	90.2	53.9	89.7
Cascaded Multi_task										
	DQPSA [36]	82.3	82.1	82.2	76.8	74.6	76.7	90.7	56.1	<u>91.1</u>
	AoM [52]	83.1	82.8	82.9	80.6	76.1	77.1	91.1	<u>56.3</u>	90.9
	Crisis-DIAS [2]	84.0	83.7	83.9	77.5	75.7	77.9	90.4	55.6	90.9
	JML [17]	82.5	81.9	82.2	81.4	78.2	80.9	<u>91.2</u>	51.9	89.9
	BiMCF (ours)	87.1	87.1	87.1	84.2	78.2	84.3	93.4	57.0	93.7

Table 2: Task sequence analysis across Acc (%) metric. The task sequence ‘1->2->3’ and ‘3->2->1’ are equivalent due to the bidirectional propagation of our BiMCF.

Task Sequence	Task1	Task2	Task3
1->2->3 3->2->1	(ours) 87.1±0.3	84.2±0.4	93.4±0.3
1->3->2 2->3->1	85.9±0.4	80.2±0.4	90.7±0.2
2->1->3 3->1->2	84.9±0.3	81.8±0.5	91.3±0.3

locating helpless features. Looking from left to right in Figure 3, we can observe that the forward and backward gates have their respective advantages and are able to attend to the image features that the other overlooks. However, they fail to locate all task-related features and attend to some irrelevant features. Fortunately, the combination of them helps alleviate these issues.

4.4 Ablation Study (RQ3)

We conduct ablation study on CrisisMMD to validate the superiority of our bidirectional cascade and display the results in Table 3.

- **Obs6: Our bidirectional propagation mechanism achieves the best performance, indicating that both our forward and backward flow are effective.** As shown in the first row in Table 3, the remarkable results demonstrate the effectiveness of our bidirectional propagation mechanism. In the next two rows, we remove the forward flow and backward flow from our model respectively. It clearly shows that there has been a certain degradation of the performance. We further carry out experiments without incorporating forward and backward flows and see that the results get worse as expected. From the above observations, we can conclude that the bidirectional propagation mechanism alleviates errors accumulated in one direction and allows these subtasks to exchange information more effectively, thus leading to further performance improvement.

4.5 Robustness Analysis (RQ4)

Table 4 presents the performance on the three tasks when employing different text and image backbones.

- **Obs7: Our proposed BiMCF demonstrates superior robustness.** From Table 4, we can clearly observe that our approach can consistently achieve excellent performance with the combinations of different text and image backbones. When we utilize more powerful feature extractors, our model can obtain better

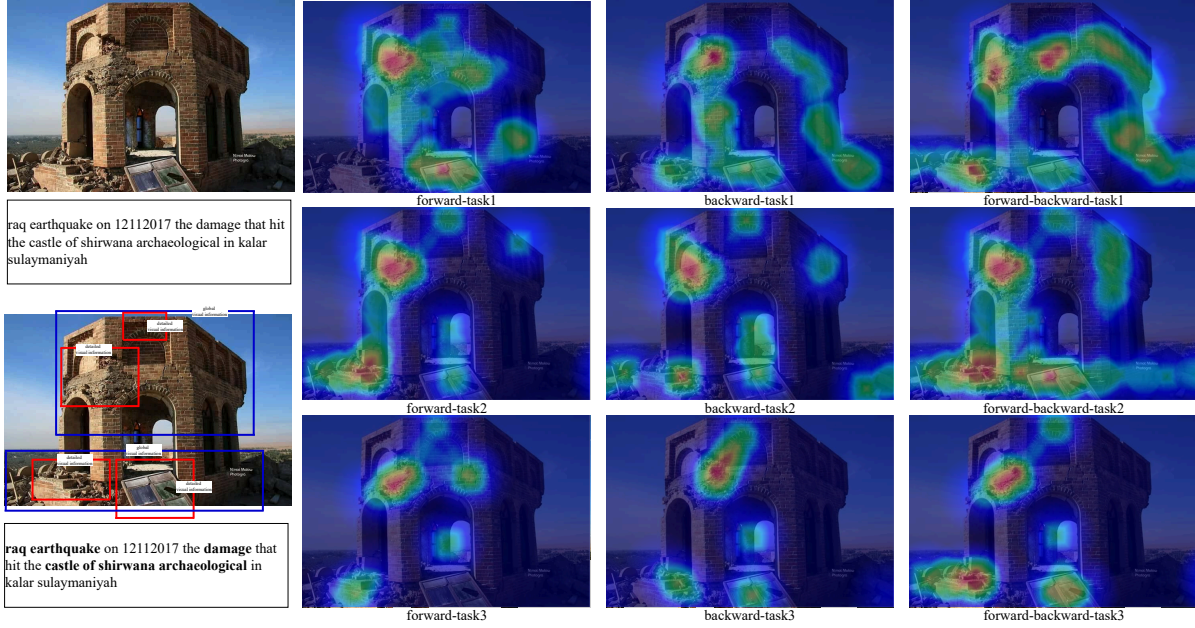


Figure 3: Visualization of the attended image information in the three subtasks, considering different pass mechanisms: forward pass only, backward pass only and forward & backward pass. Given a text-image pair, the vertical arrangement symbolizes the attended image information by different tasks after passing through the cascaded gates. The horizontal arrangement highlights the comparison of attended information among the three propagation mechanisms.

Table 3: Ablation study. The notation ‘-forward flow’ means removing forward flow from our BiMCF model, and so on.

Model design	Task 1			Task 2			Task 3		
	Acc(%)	M-F1(%)	W-F1(%)	Acc(%)	M-F1(%)	W-F1(%)	Acc(%)	M-F1(%)	W-F1(%)
BiMCF	87.1	87.1	87.1	84.2	78.2	84.3	93.4	57.0	93.7
-forward flow	85.1	84.7	85.3	81.7	75.1	81.9	90.7	55.0	90.6
-backward flow	84.9	84.8	84.8	82.3	75.5	82.1	91.3	55.6	91.4
-forward & backward flow	81.2	81.3	81.2	78.4	71.9	77.9	89.7	50.8	89.1

Table 4: The performance of our model on the combination of different text and image encoders across Acc (%) metric.

Text Encoder	Image Encoder	Task1	Task2	Task3
BERT-base	ViT	87.1±0.3	84.2±0.4	93.4±0.3
XLNet [49]	ViT	88.2±0.2	84.5±0.4	93.7±0.3
RoBERTa [24]	ViT	87.6±0.1	84.2±0.3	93.1±0.2
BERT	CLIP [40]	86.8±0.3	84.4±0.3	93.0±0.4
BERT	Swin-T [25]	87.3±0.5	84.7±0.3	93.9±0.3

results than those reported in Table 1. This observation indicates that our proposed BiMCF can effectively adapts to various backbones without compromising its accuracy.

5 Conclusion

This work proposes the bidirectional multi-task cascaded multi-modal fusion approach towards joint multimodal damage analysis. Based on this, we introduce a bidirectional multi-task cascaded

framework, which ranks the three related subtasks based on the increasing granularity. Through incorporating forward and backward flows, this framework propagates information in dual directions to alleviate errors accumulated in one direction. Meanwhile, the bidirectional flow effectively exchanges information from related tasks to enhance the prediction for the target subtask. Considering that different tasks attend to different image information, we further present a unified multimodal fusion framework to effectively integrate information from both textual and visual modalities. We evaluate this approach on the three subtasks of damage analysis, and obtain superior results across all the baselines.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 62106204), the Natural Science Foundation of Sichuan (No. 2025YFHZ0124), the Natural Science Foundation of XinjiangUygur Autonomous (No. 2024D01A14), the Frontier Cross Innovation Team Project of Southwest Jiaotong University (YH1500112432297), and the Engineering Research Center of Sustainable Urban Intelligent Transportation, Ministry of Education.

References

- [1] Mahdi Abavisani, Liwei Wu, Shengli Hu, Joel R. Tetreault, and Alejandro Jaimes. 2020. Multimodal Categorization of Crisis Events in Social Media. In *CVPR*. Computer Vision Foundation / IEEE, 14667–14677.
- [2] Mansi Agarwal, Maitree Leekha, Ramit Sawhney, and Rajiv Ratn Shah. 2020. Crisis-DIAS: Towards Multimodal Damage Analysis - Deployment, Challenges and Assessment. In *AAAI*. AAAI Press, 346–353.
- [3] Kashif Ahmad, Konstantin Pogorelov, Michael Riegler, Nicola Conci, and Pål Halvorsen. 2019. Social media and satellites - Disaster event detection, linking and summarization. *Multim. Tools Appl.* 78, 3 (2019), 2837–2875.
- [4] Firoj Alam, Ferda Ofli, and Muhammad Imran. 2018. CrisisMMD: Multimodal Twitter Datasets from Natural Disasters. In *ICWSM*. AAAI Press, 465–473.
- [5] Firoj Alam, Ferda Ofli, and Muhammad Imran. 2019. CrisisDPS: Crisis Data Processing Services. In *ISCRAM*. ISCRAM Association.
- [6] Mohammad Arif Ul Alam. 2022. College Student Retention Risk Analysis from Educational Database Using Multi-Task Multi-Modal Neural Fusion. In *AAAI*. 12689–12697.
- [7] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. 2023. Qwen-VL: A Frontier Large Vision-Language Model with Versatile Abilities. *CoRR* abs/2308.12966 (2023).
- [8] Mohammad Basit, Bashir Alam, Zubaida Fatima, and Salman Shaikh. 2023. Natural Disaster Tweets Classification Using Multimodal Data. In *EMNLP*. 7584–7594.
- [9] Yuwei Cao, Hao Peng, Zhengtao Yu, and Philip S. Yu. 2024. Hierarchical and Incremental Structural Entropy Minimization for Unsupervised Social Event Detection. In *AAAI*. AAAI Press, 8255–8264.
- [10] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *NAACL*. 4171–4186.
- [11] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelley, Jakob Uszkoreit, and Neil Houlsby. 2021. An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *ICLR*.
- [12] Junfeng Fang, Zac Bi, Ruipeng Wang, Houcheng Jiang, Yuan Gao, Kun Wang, An Zhang, Jie Shi, Xiang Wang, and Tat-Seng Chua. 2024. Towards Neuron Attributions in Multi-Modal Large Language Models. In *NeurIPS*.
- [13] Jinghan He, Kuan Zhu, Haiyun Guo, Junfeng Fang, Zhenglin Hua, Yuheng Jia, Ming Tang, Tat-Seng Chua, and Jinqiao Wang. 2024. Cracking the Code of Hallucination in LLMs with Vision-aware Head Divergence. *CoRR* abs/2412.13949 (2024). doi:10.48550/ARXIV.2412.13949
- [14] Ruidan He, Wee Sun Lee, Hwee Tou Ng, and Daniel Dahlmeier. [n. d.]. An Interactive Multi-Task Learning Network for End-to-End Aspect-Based Sentiment Analysis. In *ACL*. 504–515.
- [15] Ahmed Hendawy, Jan Peters, and Carlo D'Eramo. 2024. Multi-Task Reinforcement Learning with Mixture of Orthogonal Experts. In *ICLR*. OpenReview.net.
- [16] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *ICLR*. OpenReview.net.
- [17] Xincheng Ju, Dong Zhang, Rong Xiao, Junhui Li, Shoushan Li, Min Zhang, and Guodong Zhou. 2021. Joint Multi-modal Aspect-Sentiment Analysis with Auxiliary Cross-modal Relation Detection. In *EMNLP*. 4395–4405.
- [18] Wonjae Kim, Bokyung Son, and Ildoo Kim. 2021. ViLT: Vision-and-Language Transformer Without Convolution or Region Supervision. In *ICML*, Marina Meila and Tong Zhang (Eds.), Vol. 139. 5583–5594.
- [19] Xin Li, Lidong Bing, Wai Lam, and Bei Shi. 2018. Transformation Networks for Target-Oriented Sentiment Classification. In *ACL*. 946–956.
- [20] Kukun Li, Doina Caragea, Huaiyu Zhang, and Muhammad Imran. 2018. Localizing and Quantifying Damage in Social Media Images. In *IEEE/ACM*. 194–201.
- [21] Yangyang Li, Jun Li, Hao Jin, and Liang Peng. 2021. Focusing Attention across Multiple Images for Multimodal Event Detection. In *ACM MM*. 74:1–74:6.
- [22] Tao Liang, Guosheng Lin, Lei Feng, Yan Zhang, and Fengmao Lv. 2021. Attention is not Enough: Mitigating the Distribution Discrepancy in Asynchronous Multimodal Sequence Fusion. In *ICCV*. IEEE, 8128–8136.
- [23] Tao Liang, Guosheng Lin, Mingyang Wan, Tianrui Li, Guojun Ma, and Fengmao Lv. 2022. Expanding Large Pre-trained Unimodal Models with Multimodal Information Injection for Image-Text Multimodal Classification. In *CVPR*. 15471–15480.
- [24] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. RoBERTa: A Robustly Optimized BERT Pretraining Approach. *CoRR* abs/1907.11692 (2019).
- [25] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. 2021. Swin Transformer: Hierarchical Vision Transformer using Shifted Windows. In *ICCV*. IEEE, 9992–10002.
- [26] Dehong Ma, Sujian Li, and Houfeng Wang. 2018. Joint Learning for Targeted Sentiment Analysis. In *EMNLP*, Ellen Riloff, David Chiang, Julia Hockenmaier, and Jun'ichi Tsujii (Eds.). 4737–4742.
- [27] Krishanu Maity, Prince Jha, Sriparna Saha, and Pushpak Bhattacharyya. 2022. A Multitask Framework for Sentiment, Emotion and Sarcasm aware Cyberbullying Detection from Multi-modal Code-Mixed Memes. In *SIGIR*. 1739–1749.
- [28] Bishwas Mandal, Sarthak Khanal, and Doina Caragea. 2024. Contrastive Learning for Multimodal Classification of Crisis related Tweets. In *WWW*. ACM, 4555–4564.
- [29] Yue Mao, Yi Shen, Chao Yu, and Longjun Cai. 2021. A Joint Training Dual-MRC Framework for Aspect Based Sentiment Analysis. In *AAAI*. AAAI Press, 13543–13551.
- [30] Yiqiao Mao, Xiaoqiang Yan, Zirui Hu, Xuguang Zhang, Yangdong Ye, and Hui Yu. 2024. Memory-aware continual learning with multi-modal social media streams for unsupervised disaster classification. *Advanced Engineering Informatics* 62 (2024), 102654.
- [31] Subhojyoti Mukherjee, Qiaomin Xie, Josiah Hanna, and Robert D. Nowak. 2023. Multi-task Representation Learning for Pure Exploration in Bilinear Bandits. In *NeurIPS*.
- [32] Kai Nakamura, Sharon Levy, and William Yang Wang. 2020. Fakeddit: A New Multimodal Benchmark Dataset for Fine-grained Fake News Detection. In *LREC*. European Language Resources Association, 6149–6157.
- [33] Ganesh Nalluru, Rahul Pandey, and Hemant Purohit. 2019. Relevancy Classification of Multimodal Social Media Streams for Emergency Services. In *SMART-COMP*. 121–125.
- [34] Thi Huyen Nguyen and Koustav Rudra. 2024. Human vs ChatGPT: Effect of Data Annotation in Interpretable Crisis-Related Microblog Classification. In *WWW May 13-17, 2024*. ACM, 4534–4543.
- [35] Marco Aurelio Deoldoto Paulino, Yandre M. G. Costa, and Valéria Delisandra Feltrim. 2022. Evaluating multimodal strategies for multi-label movie genre classification. In *IWSSIP*. 1–4.
- [36] Tianshuo Peng, Zuchao Li, Ping Wang, Lefei Zhang, and Hai Zhao. 2024. A Novel Energy Based Model Mechanism for Multi-Modal Aspect-Based Sentiment Analysis. In *AAAI*. AAAI Press, 18869–18878.
- [37] Georgios Petkos, Symeon Papadopoulos, and Yiannis Kompatsiaris. 2012. Social event detection using multimodal clustering and integrating supervisory signals. In *ICMR*. 23.
- [38] Raj Pranesh. 2022. Exploring Multimodal Features and Fusion Strategies for Analyzing Disaster Tweets. In *COLING*. 62–68.
- [39] Zitai Qiu, Congbo Ma, Jia Wu, and Jian Yang. 2024. An Efficient Automatic Meta-Path Selection for Social Event Detection via Hyperbolic Space. In *WWW*. ACM, 2519–2529.
- [40] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *ICML*, Vol. 139. PMLR, 8748–8763.
- [41] Haosen Shi, Shen Ren, Tianwei Zhang, and Sinno Jialin Pan. 2023. Deep Multitask Learning with Progressive Parameter Sharing. In *ICCV*. IEEE, 19867–19878.
- [42] Madichetty Sreenivasulu and M. Sridevi. 2019. Detecting Informative Tweets during Disaster using Deep Neural Networks. In *COMSNETS*. 709–713.
- [43] Kevin Stowe, Michael J. Paul, Martha Palmer, Leysia Palen, and Ken Anderson. 2016. Identifying and Categorizing Disaster-Related Tweets. In *EMNLP*. 1–6.
- [44] Weijie Su, Xizhou Zhu, Yue Cao, Bin Li, Lewei Lu, Furu Wei, and Jifeng Dai. 2020. VL-BERT: Pre-training of Generic Visual-Linguistic Representations. In *ICLR*. OpenReview.net.
- [45] Kaiwen Tan, Weixian Huang, Xiaofeng Liu, Jinlong Hu, and Shoubin Dong. 2022. A multi-modal fusion framework based on multi-task correlation learning for cancer prognosis prediction. *Artif. Intell. Medicine* 126 (2022), 102260.
- [46] Yi Tay, Luu Anh Tuan, and Siu Cheung Hui. 2018. Learning to Attend via Word-Aspect Associative Fusion for Aspect-Based Sentiment Analysis. In *AAAI*. 5956–5963.
- [47] Wei Han Wang, Qingsong Lv, Wenmeng Yu, Wenyi Hong, Ji Qi, Yan Wang, Junhui Ji, Zhuoyi Yang, Lei Zhao, Xixuan Song, Jiazheng Xu, Bin Xu, Juanzi Li, Yuxiao Dong, Ming Ding, and Jie Tang. 2023. CogVLM: Visual Expert for Pretrained Language Models. *CoRR* abs/2311.03079 (2023).
- [48] Xuehua Wu, Jin Mao, Hao Xie, and Gang Li. 2022. Identifying humanitarian information for emergency response by modeling the correlation and independence between text and images. *Inf. Process. Manag.* 59, 4 (2022), 102977.
- [49] Zhilin Yang, Zihang Dai, Yiming Yang, Jaime G. Carbonell, Ruslan Salakhutdinov, and Quoc V. Le. 2019. XLNet: Generalized Autoregressive Pretraining for Language Understanding. In *NeurIPS*. 5754–5764.
- [50] Yayu Zhang, Yuhua Qian, Guoshuai Ma, Keyin Zheng, Guoqing Liu, and Qingfu Zhang. 2024. Learning Multi-Task Sparse Representation Based on Fisher Information. In *AAAI*. AAAI Press, 16899–16907.
- [51] Yu Zhang and Qiang Yang. 2022. A Survey on Multi-Task Learning. *TKDE*. 34, 12 (2022), 5586–5609.
- [52] Ru Zhou, Wenya Guo, Xumeng Liu, Shenglong Yu, Ying Zhang, and Xiaojie Yuan. 2023. AoM: Detecting Aspect-oriented Information for Multimodal Aspect-Based Sentiment Analysis. In *ACL*. 8184–8196.
- [53] Zhiqiang Zou, Hongyu Gan, Quanying Huang, Tianhui Cai, and Kai Cao. 2021. Disaster Image Classification by Fusing Multimodal Social Media Data. *ISPRS Int. J. Geo Inf.* 10, 10 (2021), 636.

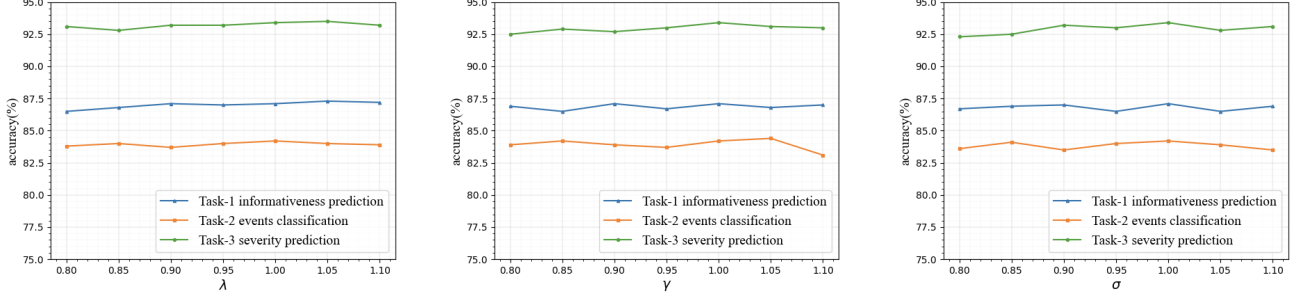


Figure 4: Sensitivity analysis for the hyper-parameters (λ , γ and σ) in terms of classification accuracy. We conduct the experiments by changing the value of one hyper-parameter within a certain range, while setting the values of other hyper-parameters to a constant 1. All the three subtasks are included into the analysis experiment.

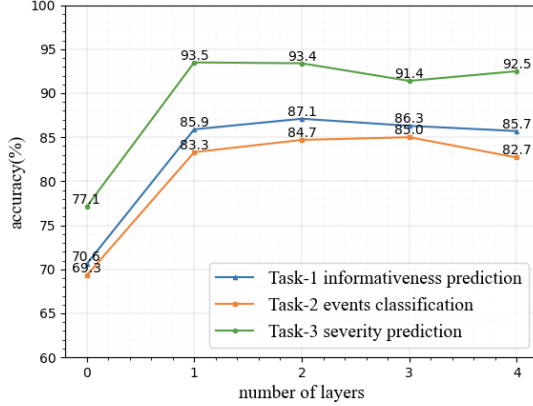


Figure 5: Analysis on the layer number of bidirectional flow in terms of classification accuracy.

Table 5: Fakeddit-mini dataset statistics. ‘True’ denotes true news, ‘Fake-True’ denotes fake news with true text, others denote the specific categories of fake news with false text.

Sample Types	# of samples
True	8000
Fake-True	2000
Satire/Parody	2000
Misleading Content	2000
Imposter Content	2000
False Connection	2000
Manipulated Content	4000
Total	20000

A Supplementary Experimental Results

A.1 Hyper-parameter Sensitivity

In this section, we analyze the sensitivity of our BiMCF on the CrisisMMD to hyper-parameters and display the result in Figure 4.

- **Obs1a: The performance of our BiMCF is not sensitive to the variations of hyper-parameters.** We examine the impact

on the identification of the three subtasks when changing the trade-off hyper-parameter λ for L_{D_1} , γ for L_{D_2} and σ for L_{D_3} respectively. Taking the experiment on α in Figure 4 as an example, we change the value of λ within a certain range and keep the value of other hyper-parameters as the constant 1. We can easily find that the three curves, representing the classification accuracy of the three subtasks, exhibits minimal fluctuations and keep smooth, which means that our method is not sensitive to λ . Similarly, we can also find that our BiMCF is not sensitive to γ and σ as well. Therefore, the above analysis fully demonstrates the robustness of our proposed approach. Moreover, in all experiments on the CrisisMMD benchmark, we set the values of hyperparameters λ , γ , and σ to 1.

A.2 Layer Analysis

In this section, we comprehensively investigate how the number of bidirectional flows influences the performance of our proposed approach, and the corresponding results on the three subtasks are shown in Figure 5.

- **Obs2a: When the number of the adopted bidirectional flow is set to 2, the model achieves the optimal overall performance.** From Figure 5, we can observe that the performance of our approach is definitely improved when the bidirectional propagation flow is introduced into our model. This improvement suggests that our bidirectional flow facilitates the feature extraction process and enhances the fusion of image and text modalities, thus leading to better overall performance. With the increase of the layer number, the performance of the model gradually improves at first, and then begins to decrease after reaching the best. Therefore, in our experiment, we adopt two bidirectional flows, which enables our proposed approach to achieve the optimal performance overall.

B Experiment on Another Dataset

In order to demonstrate the generalizability of our proposed BiMCF, we further showcase its application in the field of fake news detection and demonstrate its performance on the Fakeddit [32] benchmark. Moreover, in this experiment, we adopt the same settings as in the experiments about the CrisisMMD.

Table 6: Performance comparison with baselines on Fakeddit-mini. ‘Single_task’ denotes treating each subtask independently. ‘Parallel multi_task’ and ‘Cascaded multi_task’ address all three subtasks simultaneously. However, the former doesn’t consider the inherent dependencies between subtasks, while the latter does. The best results are in bold, and the second-best results are underlined. Metrics include Acc: classification accuracy and F1: F1-score.

Category	Method	Task 1		Task 2		Task 3	
		Acc(%)	F1(%)	Acc(%)	F1(%)	Acc(%)	F1(%)
Single_task	Cross-Attention [1]	90.2	90.1	<u>87.4</u>	<u>87.1</u>	81.3	80.8
	VLBert [44]	89.4	89.2	87.1	86.7	81.2	80.3
	ViLT [18]	88.7	88.8	86.3	86.1	80.8	80.3
	Qwen-VL [7]	85.3	85.2	83.2	83.0	75.5	75.2
	Qwen-VL+LoRA [16]	87.1	87.2	85.4	85.7	79.3	78.6
	CogVLM [47]	86.9	87.0	84.9	84.7	77.6	77.1
	CogVLM+LoRA	88.4	88.5	86.7	86.6	80.3	79.7
Parallel Multi_task							
Multi_task	Cross-Attention	86.5	86.6	83.2	82.9	80.1	79.7
	VLBert	87.8	87.7	84.6	84.7	79.4	78.8
	ViLT	85.7	85.9	83.7	83.6	77.2	76.6
	Qwen-VL	82.2	82.1	83.6	83.7	74.7	74.3
	Qwen-VL+LoRA	84.7	84.8	84.9	84.7	77.4	76.8
	CogVLM	83.8	83.9	82.4	82.2	75.1	74.5
	CogVLM+LoRA	85.6	85.5	83.9	83.7	78.4	78.1
	IMN [14]	86.7	86.7	84.1	83.8	80.6	80.2
Cascaded Multi_task							
	DQPSA [36]	87.8	87.6	83.2	83.4	78.4	78.1
	AoM [52]	<u>91.2</u>	<u>91.1</u>	84.7	84.9	80.8	80.1
	Crisis-DIAS [2]	89.7	89.8	86.6	86.4	<u>81.4</u>	<u>80.9</u>
	JML [17]	90.5	90.3	85.6	85.2	80.5	79.8
	BiMCF (ours)	91.8	92.2	88.1	87.8	83.4	82.3

B.1 Implementation Details

Dataset. The Fakeddit [32] is a multimodal dataset specifically designed for fake news detection, and it also exhibits cascading relationships between the three subtasks. Given the large scale of the Fakeddit (over 1 million samples) and our limited time, we randomly sampled 20,000 samples from the Fakeddit and reconstruct a dataset named Fakeddit-mini. Additionally, we ensured that the class distribution within this dataset is relatively balanced to provide a comprehensive evaluation of our method’s performance.. And the labels of each sample remain in their original form. Specifically, each sample has three labels. The first label is marked as fake or true to determine whether the sample is fake news (task 1). The second label is marked as fake-true or fake-fake to determine whether the sample is fake with true text or the sample is fake with false text (task 2). The third label is marked as satire/parody, misleading content, imposter content, false connection or manipulated content to categorize different types of fake samples with false text (task 3). Similar to the crisisMMD dataset, the three tasks exhibit cascading relationships and the class label of the latter task

depends on that of the former task. The detailed information of Fakeddit-mini is displayed in Table 5.

B.2 Performance Comparison

We compare our BiMCF with state-of-the-art multi-task or single-task methods. And Table 6 presents the performance of our method on the Fakeddit-mini dataset across classification accuracy and F1 score. And we can draw the following observations.

Obs3a: Our proposed BiMCF achieves state-of-the-art results across all the adopted baselines, demonstrating satisfactory performance improvements compare to both single_task approaches and multi_task approaches. For example, compared to Crisis - DIAS method, which achieves second-best performance on task3, our BiMCF improves the classification accuracy by 2%.

Obs4a: Our proposed BiMCF exhibits superior capability in fake news detection compared to MLLMs. Whether using fine-tuning or text prompts for disaster analysis with MLLMs, neither approach fully explores the cascading relationships across subtasks. Additionally, the use of text prompts lacks the guidance of damage-specific knowledges related to fake news. Therefore, they fail to

Table 7: Ablation study. The notation ‘-forward flow’ means removing forward flow from our BiMCF model, and so on.

<i>Model design</i>	<i>Task 1</i>		<i>Task 2</i>		<i>Task 3</i>	
	<i>Acc(%)</i>	<i>F1(%)</i>	<i>Acc(%)</i>	<i>F1(%)</i>	<i>Acc(%)</i>	<i>F1(%)</i>
BiMCF	91.8	92.2	88.1	87.8	83.4	82.3
-forward flow	89.7	89.5	87.3	87.1	82.1	81.3
-backward flow	87.4	87.6	84.2	84.1	80.4	79.6
-forward & backward flow	82.7	82.9	79.3	79.5	77.6	77.1

Table 8: The performance of our model on the combination of different text and image encoders across Acc (%) metric.

<i>Text Encoder</i>	<i>Image Encoder</i>	<i>Task1</i>	<i>Task2</i>	<i>Task3</i>
BERT-base	ViT	91.8±0.2	88.1±0.3	83.4±0.3
XLNet [49]	ViT	92.3±0.4	88.7±0.2	84.1±0.2
RoBERTa [24]	ViT	92.5±0.3	88.5±0.3	83.7±0.3
BERT	CLIP [40]	92.7±0.2	89.1±0.4	84.1±0.3
BERT	Swin-T [25]	92.2±0.3	88.4±0.2	83.7±0.4

ourperform the performance of our model.

Obs5a: The excellent performance of our model on both the Fakeddit and CrisisMMD datasets clearly demonstrates its generalizability to different domains.

B.3 Ablation Study

In order to validate the superiority of our bidirectional cascade, we conduct comprehensive ablation study on the Fakeddit-mini benchmark and display the results in Table 7.

obs6a: Our bidirectional propagation mechanism performs

better than the unidirectional propagation mechanism. And with the introduction of the bidirectional flow, the classification accuracy of task 1 is improved by 9.1% and achieves 91.8%. In Table 7, the introduction of forward flow and backward flow can improve the model’s performance respectively, as they explore the cascading relationships between subtasks. Based on this, our BiMCF can further enhance the performance, as the bidirectional flow effectively suppresses the propagation of negative messages and allows subtasks to exchange information more effectively.

B.4 Robustness Analysis

Table 8 presents the performance on the three tasks in the Fakeddit-mini dataset when employing different text and image backbones. We can draw the following observation.

Obs7a: Our proposed BiMCF demonstrates superior robustness on Fakeddit-mini dataset. From Table 8, we can clearly observe that our approach can consistently achieve excellent performance with the combinations of different text and image backbones. When we utilize more powerful feature extractors, our model can obtain better results. This observation indicates that our proposed BiMCF can effectively adapts to various backbones without compromising its accuracy.